

Answer Key — Biology Grade 11

Sub-Strand 3.1: Reproduction in Animals • Lesson 4: Hormonal Conductors: How Pituitary and Ovarian Hormones Regulate the Menstrual Cycle and Egg Release

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	B	B	D	B	C	C

USE AFTER: OBSERVE PHASE — Observe: tracing FSH, LH, oestrogen and progesterone on the hormone charts and graph worksheets

Q1. On a normal 28-day menstrual cycle graph, a sudden sharp rise (surge) in one hormone around Day 14 is followed soon after by ovulation. Which hormone surge triggers the release of the egg from the ovary?

- A. Oestrogen
- B. Follicle stimulating hormone (FSH)
- C. Luteinising hormone (LH)**
- D. Progesterone

Correct answer: C. The LH surge around Day 14 triggers ovulation. FSH acts earlier to stimulate follicle growth, and progesterone rises only after ovulation.

USE AFTER: OBSERVE PHASE — Observe: observation table (ovary event: corpus luteum; uterine lining: maintained/breakdown)

Q2. After ovulation, the empty follicle becomes the corpus luteum. Which hormone does the corpus luteum mainly release to keep the uterine lining thick and ready for a fertilised egg?

- A. Luteinising hormone (LH)
- B. Progesterone**
- C. Oestrogen
- D. Follicle stimulating hormone (FSH)

Correct answer: B. The corpus luteum mainly secretes progesterone, which maintains the thickened uterine lining in the second half of the cycle. Oestrogen's main role is the repair and thickening of the lining before ovulation.

USE AFTER: OBSERVE PHASE — Observe: 'imbalance' scenario cards (absent LH surge with no ovulation)

Q3. A woman's hormone graph shows normal FSH and oestrogen, but there is no LH surge in the middle of the cycle. What is the most likely effect?

- A. Ovulation takes place but progesterone rises too early
- B. The egg is not released because ovulation does not take place**
- C. Several eggs are released at the same time
- D. The uterine lining thickens twice in one cycle

Correct answer: B. The LH surge triggers ovulation, so without it the follicle does not release the egg. This is the 'absent LH surge, no ovulation' scenario, where fertility treatment may help.

USE AFTER: EXPLAIN PHASE — Explain: building the hormonal flowchart/concept map and labelling negative feedback arrows

Q4. In the second half of the cycle, oestrogen and progesterone levels are high. What do these high levels do to the pituitary gland, as an example of negative feedback?

- A. They make it release the egg directly
- B. They stimulate it to release even more FSH and LH
- C. They have no effect on it
- D. They inhibit (reduce) the release of FSH and LH**

Correct answer: D. High progesterone and oestrogen tell the pituitary to reduce FSH and LH, like a thermostat switching off heating. This stops new follicles developing while the lining is maintained.

USE AFTER: EXPLAIN PHASE — Explain: building the hormonal flowchart/concept map and labelling positive feedback arrows

Q5. Just before ovulation, oestrogen from the growing follicle reaches a high level. What does this high oestrogen do to LH release, as an example of positive feedback?

- A. It stops all LH release until menstruation
- B. It triggers the LH surge**
- C. It causes menstruation to start
- D. It turns the follicle into the uterine lining

Correct answer: B. High oestrogen switches from inhibiting LH to triggering the LH surge. This is positive feedback, and the surge then causes ovulation.

USE AFTER: DQB PHASE — Observe/DQB: link between imbalance scenario cards and why IVF patients receive FSH injections (DQB question 'How do IVF clinics in Nairobi use hormones?')

Q6. A woman at a clinic in Nairobi is given FSH injections as part of IVF treatment. Why is FSH given?

- A. To make the uterus grow a baby outside the body
- B. To replace progesterone after ovulation
- C. To stimulate several follicles to develop, so more than one egg can be collected**
- D. To stop menstruation permanently

Correct answer: C. FSH stimulates follicle growth. Injections make several follicles develop, whereas normally only one egg is released each cycle.

USE AFTER: MODEL BUILDING PHASE — Model: adding the 'Hormonal control of gamete release' layer and correcting the IVF 'test tube baby' misconception

Q7. Many people believe IVF babies grow wholly in a test tube. Which statement is scientifically correct?

- A. IVF babies need no hormones at any stage of development
- B. The baby grows in the test tube for nine months, then is placed in the mother
- C. Only fertilisation happens in the laboratory; the baby develops in the mother's uterus**
- D. The baby develops in the ovary after fertilisation in the laboratory

Correct answer: C. In IVF, fertilisation takes place in the laboratory, and the embryo is then placed in the uterus, where development takes place. Hormones still play a role in preparing and maintaining the uterus.